Titanic Survival Classification Using Supervised ML

This notebook aims to **predict** whether a passenger survived the Titanic disaster. We'll use three classification models:

1. **Logistic Regression** (baseline, more interpretable)
2. **Random Forest** (generally good predictive power)
3. **XGBoost** (often among the top-performing classifiers)

We'll then compare their performance, select the best model, and summarize key insights. https://www.kaggle.com/datasets/luvippp/titanic-updated-for-learning/data **It's my dataset having a simple update on survival rate among passengers because original was incoherent with no men surviving** however, the reality show another repartition https://titanicfacts.net/titanic-survivors/#:~:text=31.6%25%20%E2%80%93%20the%20percentage%20of%20people,spaces%20available%20on%20the%20lifeboats.

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1. Main Objective

Objective: Build a classifier to predict passenger survival.
Business Benefit: Demonstrate end-to-end machine learning workflow (cleaning, feature engineering, modeling, and interpretation) applicable to any classification problem in a business context.

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2. About the Dataset

We'll use the Titanic dataset, which includes:

- **PassengerId**: Unique passenger identifier
- **Survived**: 0 = No, 1 = Yes (Target)
- **Pclass**: Passenger class (1, 2, 3)
- **Name**: Full name
- **Sex**: Gender (male/female)
- **Age**: Passenger age in years
- **SibSp**: # of siblings/spouses aboard
- **Parch**: # of parents/children aboard
- **Ticket**: Ticket number
- **Fare**: Ticket fare
- **Cabin**: Cabin number
- **Embarked**: Port of embarkation (C/Q/S)

Goal: Predict `Survived` and identify key features that drive survival.

In [48]:

```python
# import necessary modules
import pandas as pd
import re
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
warnings.filterwarnings('ignore')
```

In [49]:

```python
# Let's load the dataset
df = pd.read_csv('Titanic_updated.csv')
df.info()
df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  418 non-null    int64  
 1   Survived     418 non-null    int64  
 2   Pclass       418 non-null    int64  
 3   Name         418 non-null    object 
 4   Sex          418 non-null    object 
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64  
 7   Parch        418 non-null    int64  
 8   Ticket       418 non-null    object 
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     object 
 11  Embarked     418 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

Out[49]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	1	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

3. Exploratory Data Analysis and Cleaning

3.1 Check missing values

In [50]:

```python
print("Missing values:")
print(df.isnull().sum())
```

```
Missing values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             86
SibSp            0
Parch            0
Ticket           0
Fare             1
Cabin          327
Embarked         0
dtype: int64
```

3.2 Data correction

In [51]:

```python
# Fill missing Age with median
df.fillna({'Age':df['Age'].median()}, inplace=True)
# Fill missing Embarked with most frequent
df.fillna({'Embarked': df['Embarked'].mode()[0]}, inplace=True)
# Fill missing Fare with median
df.fillna({'Fare': df['Fare'].median()}, inplace=True)
# Create a binary feature for Cabin
df['HasCabin'] = df['Cabin'].notnull().astype(int)
#Lots of cabines are Nan because lower class didn't have assigned cabins or their cabin information was not recorded
```

3.3 Feature engineering

In [52]:

```python
# Extract title from Name (simple approach)
def get_title(name):
    match = re.search(r',\s*([^.]*)\.', name)
    return match.group(1) if match else "Unknown"

df['Title'] = df['Name'].apply(get_title)

# Group rare titles
common_titles = ["Mr", "Miss", "Mrs", "Master"]
df['Title'] = df['Title'].apply(lambda x: x if x in common_titles else "Rare")

# Drop unused columns as we updated them to be usable or removed useless info
df.drop(['PassengerId','Name','Ticket','Cabin'], axis=1, inplace=True)
```

3.4 Encode categorical values

In [53]:

```python
df = pd.get_dummies(df, columns=['Sex','Embarked','Title'], drop_first=True)

# Final columns check
print("Columns after cleaning:")
print(df.columns)
print(df.head())
print("Missing values:")
print(df.isnull().sum())
```

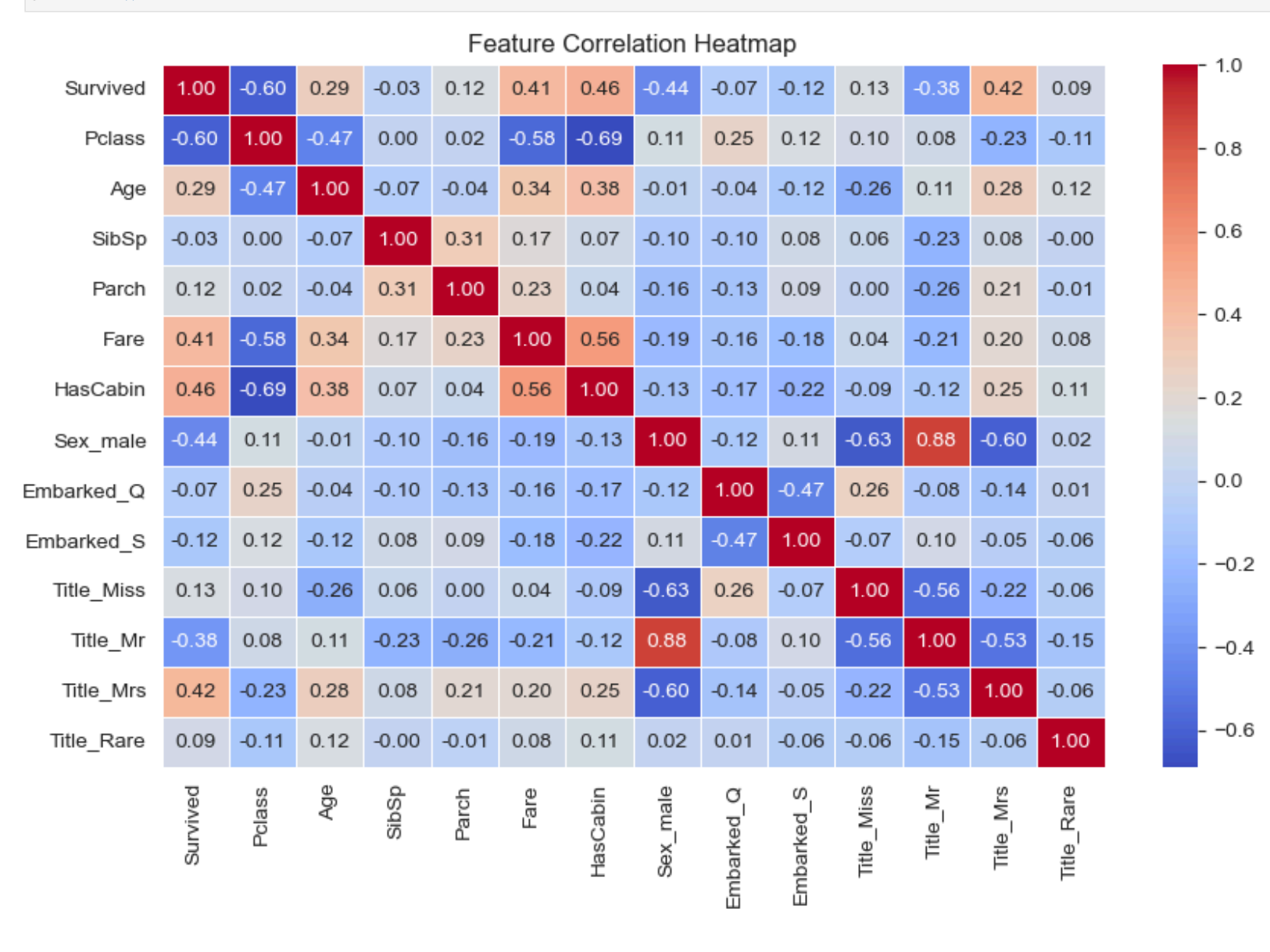
```
Columns after cleaning:
Index(['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'HasCabin',
       'Sex_male', 'Embarked_Q', 'Embarked_S', 'Title_Miss', 'Title_Mr',
       'Title_Mrs', 'Title_Rare'],
      dtype='object')
   Survived  Pclass   Age  SibSp  Parch     Fare  HasCabin  Sex_male  \
0         0       3  34.5      0      0   7.8292         0      True   
1         1       3  47.0      1      0   7.0000         0     False   
2         1       2  62.0      0      0   9.6875         0      True   
3         0       3  27.0      0      0   8.6625         0      True   
4         1       3  22.0      1      1  12.2875         0     False   

   Embarked_Q  Embarked_S  Title_Miss  Title_Mr  Title_Mrs  Title_Rare  
0        True       False       False      True      False       False  
1       False        True       False     False       True       False  
2        True       False       False      True      False       False  
3       False        True       False      True      False       False  
4       False        True       False     False       True       False  
Missing values:
Survived      0
Pclass        0
Age           0
SibSp         0
Parch         0
Fare          0
HasCabin      0
Sex_male      0
Embarked_Q    0
Embarked_S    0
Title_Miss    0
Title_Mr      0
Title_Mrs     0
Title_Rare    0
dtype: int64
```

Great ! There no missing values and our data are ready to be use for ML

3.5 Exploratory Data Analysis (EDA) & Data Consistency Checks

In [54]:

```python
# CORRELATION MATRIX (HEATMAP)
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Feature Correlation Heatmap")
plt.show()
```



	Survived	Pclass	Age	SibSp	Parch	Fare	HasCabin	Sex_male	Embarked_Q	Embarked_S	Title_Miss	Title_Mr	Title_Mrs	Title_Rare
Survived	1.00	-0.60	0.29	-0.03	0.12	0.41	0.46	-0.44	-0.07	-0.12	0.13	-0.38	0.42	0.09
Pclass	-0.60	1.00	-0.47	0.00	0.02	-0.58	-0.69	0.11	0.25	0.12	0.10	0.08	-0.23	-0.11
Age	0.29	-0.47	1.00	-0.07	-0.04	0.34	0.38	-0.01	-0.04	-0.12	-0.26	0.11	0.28	0.12
SibSp	-0.03	0.00	-0.07	1.00	0.31	0.17	0.07	-0.10	-0.10	0.08	0.06	-0.23	0.08	-0.00
Parch	0.12	0.02	-0.04	0.31	1.00	0.23	0.04	-0.16	-0.13	0.09	0.00	-0.26	0.21	-0.01
Fare	0.41	-0.58	0.34	0.17	0.23	1.00	0.56	-0.19	-0.16	-0.18	0.04	-0.21	0.20	0.08
HasCabin	0.46	-0.69	0.38	0.07	0.04	0.56	1.00	-0.13	-0.17	-0.22	-0.09	-0.12	0.25	0.11
Sex_male	-0.44	0.11	-0.01	-0.10	-0.16	-0.19	-0.13	1.00	-0.12	0.11	-0.63	0.88	-0.60	0.02
Embarked_Q	-0.07	0.25	-0.04	-0.10	-0.13	-0.16	-0.17	-0.12	1.00	-0.47	0.26	-0.08	-0.14	0.01
Embarked_S	-0.12	0.12	-0.12	0.08	0.09	-0.18	-0.22	0.11	-0.47	1.00	-0.07	0.10	-0.05	-0.06
Title_Miss	0.13	0.10	-0.26	0.06	0.00	0.04	-0.09	-0.63	0.26	-0.07	1.00	-0.56	-0.22	-0.06
Title_Mr	-0.38	0.08	0.11	-0.23	-0.26	-0.21	-0.12	0.88	-0.08	0.10	-0.56	1.00	-0.53	-0.15
Title_Mrs	0.42	-0.23	0.28	0.08	0.21	0.20	0.25	-0.60	-0.14	-0.05	-0.22	-0.53	1.00	-0.06
Title_Rare	0.09	-0.11	0.12	-0.00	-0.01	0.08	0.11	0.02	0.01	-0.06	-0.06	-0.15	-0.06	1.00

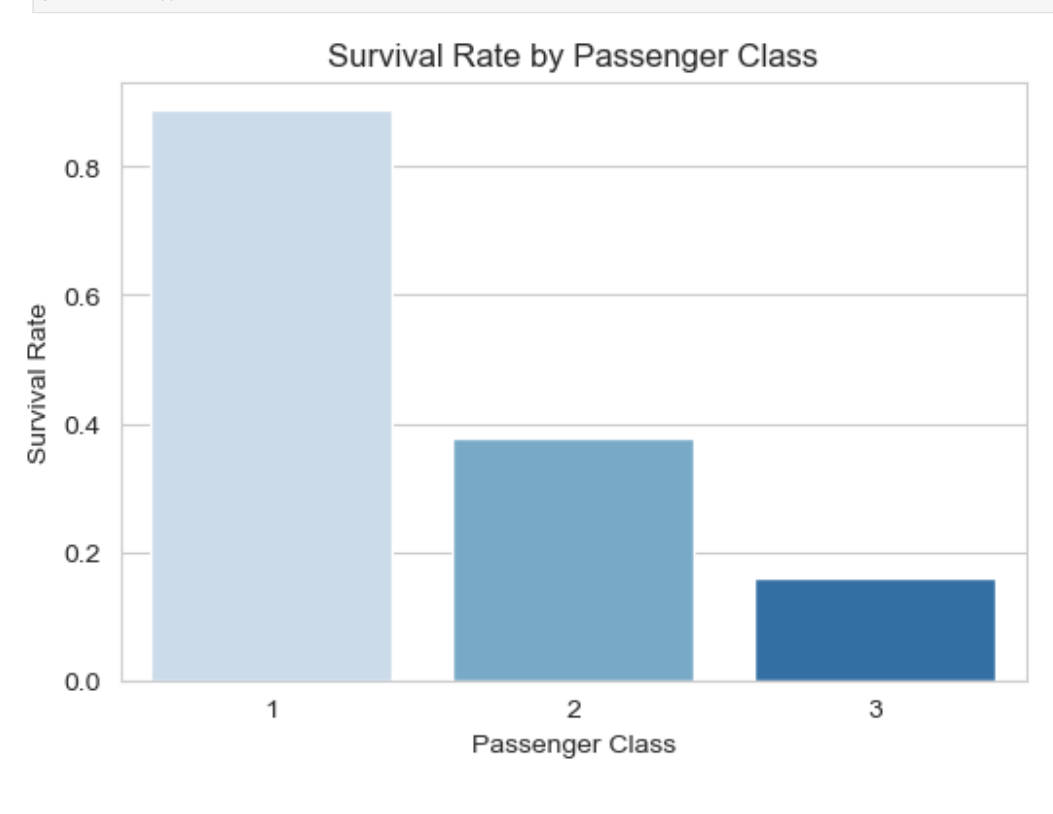
Regarding Survived, here are some notable correlations in the heatmap:

- **Pclass (-0.60): The higher the class (1st class), the greater the chances of survival. Conversely, 3rd class passengers were less likely to survive.**
- **Fare (0.41): Passengers who paid a higher price were more likely to survive, which is consistent with the correlation between class and survival.**
- **HasCabin (0.46): Having a cabin is positively correlated with survival, suggesting that better-seated passengers (often in 1st or 2nd class) were more likely to survive.**
- **Sex_male (-0.44): Being male is negatively correlated with survival, reflecting the "women and children first" rule.**
- **Title_Mr (-0.38): The title "Mr" (which mainly concerns men) is negatively correlated with survival, reinforcing the idea that men were less likely to survive.**
- **Title_Mrs (0.42) and Title_Miss (0.13): Being a woman (with the titles "Mrs" or "Miss") is positively correlated with survival, confirming that women were privileged during the evacuation.**

In summary, women, higher-class passengers, and those who paid a high price were more likely to survive, while men and lower-class passengers were less likely.

In [55]:

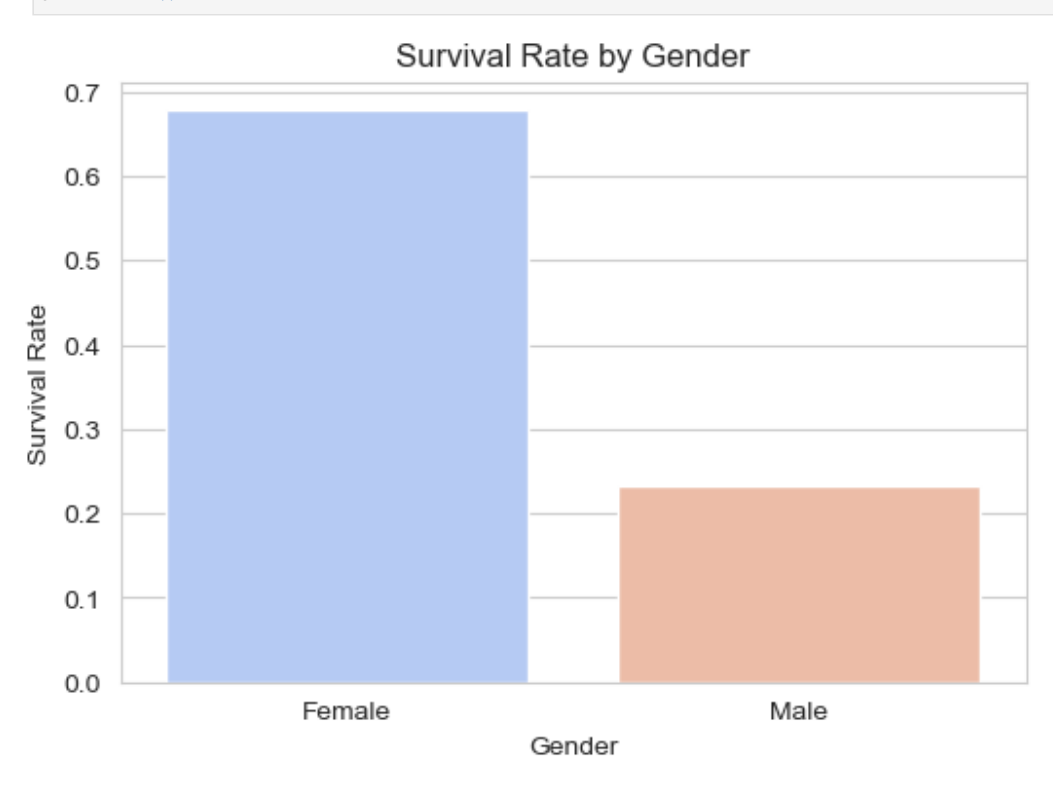
```python
# SURVIVAL RATE BY PCLASS
plt.figure(figsize=(6,4))
sns.barplot(x="Pclass", y="Survived", data=df, errorbar=None, palette="Blues")
plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate")
plt.show()
```



There no surprise since Titanic is a sort of demonstration of how wealthy are prioritized, like a parody of our society almost no 3rd class people survived.

In [56]:

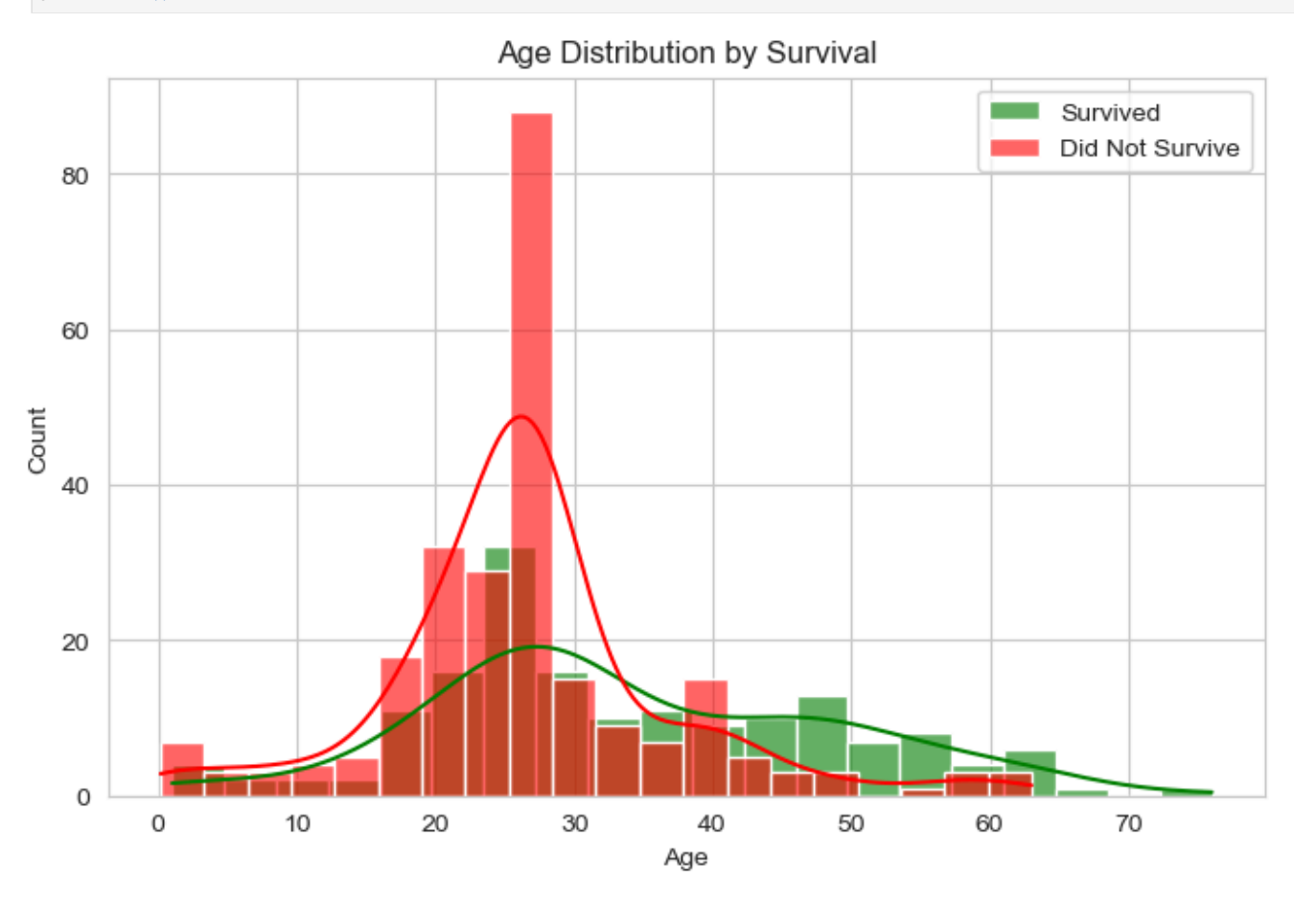
```python
# SURVIVAL RATE BY GENDER
plt.figure(figsize=(6,4))
sns.barplot(x="Sex_male", y="Survived", data=df, errorbar=None, palette="coolwarm")
plt.xticks([0,1], ['Female', 'Male'])
plt.title("Survival Rate by Gender")
plt.xlabel("Gender")
plt.ylabel("Survival Rate")
plt.show()
```



Sticking the rule "Kid and Women fisrt", this surviving rate by gender is logic and reflect reality of the catastroph

In [57]:

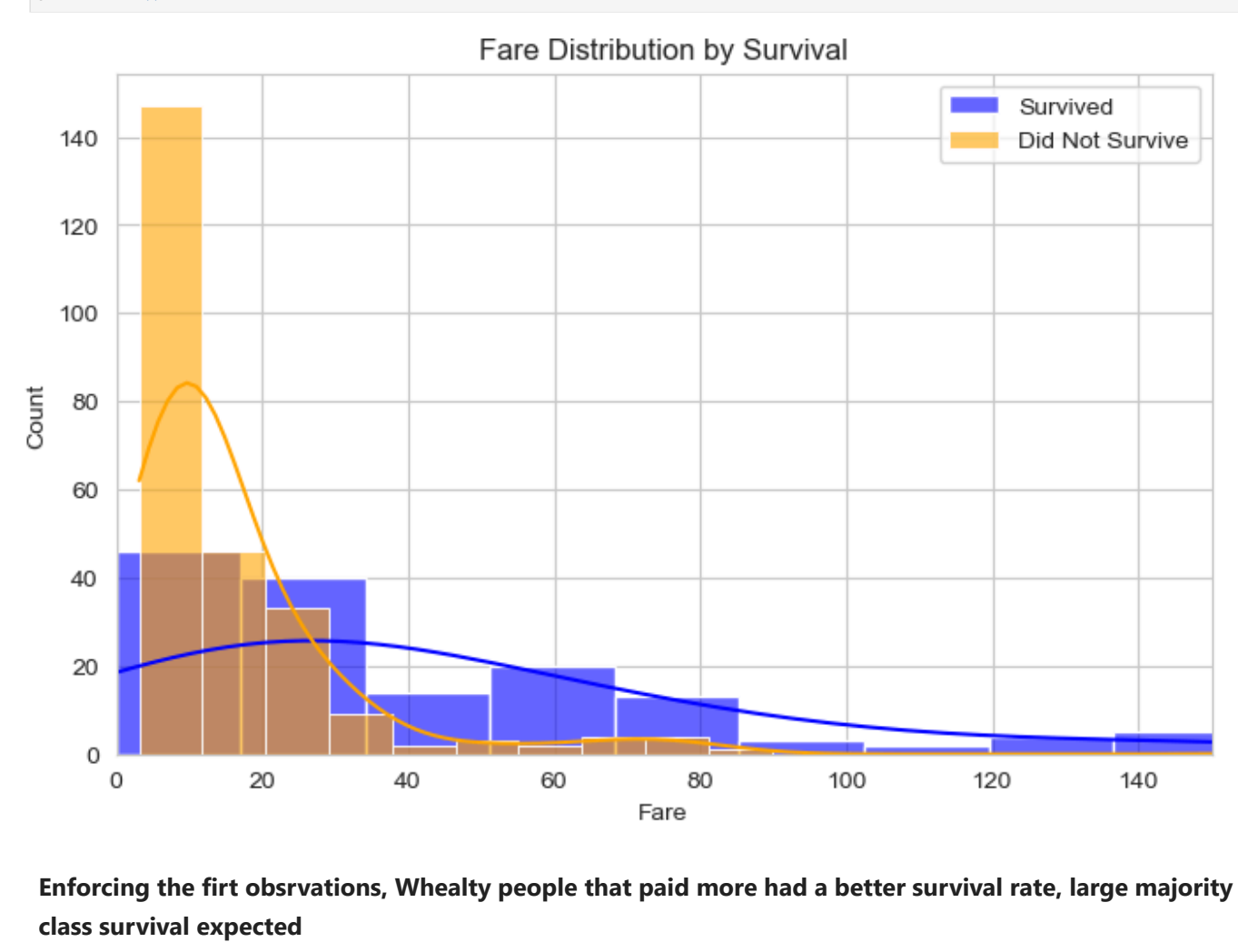
```python
# AGE DISTRIBUTION BY SURVIVAL
plt.figure(figsize=(8,5))
sns.histplot(df[df['Survived'] == 1]['Age'], bins=20, kde=True, color='green', label="Survived", alpha=0.6)
sns.histplot(df[df['Survived'] == 0]['Age'], bins=20, kde=True, color='red', label="Did Not Survive", alpha=0.6)
plt.legend()
plt.title("Age Distribution by Survival")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



Older survived more as they where the richest, a large majority of people where between 20 and 30 willing to join USA. This data makes sens

In [58]:

```python
# FARE DISTRIBUTION BY SURVIVAL
plt.figure(figsize=(8,5))
sns.histplot(df[df['Survived'] == 1]['Fare'], bins=30, kde=True, color='blue', label="Survived", alpha=0.6)
sns.histplot(df[df['Survived'] == 0]['Fare'], bins=30, kde=True, color='orange', label="Did Not Survive", alpha=0.6)
plt.legend()
plt.xlim(0, 150)  # Remove extreme outliers in visualization
plt.title("Fare Distribution by Survival")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```



Enforcing the firt obsrvations, Whealty people that paid more had a better survival rate, large majority of death where in 3 and 2 class (tickets under 40 dollars) Our data look coherent, following age, sex, class survival expected

---

4. Training Three Classifiers

We'll train:

1. **Logistic Regression**
2. **Random Forest**
3. **XGBoost**

as we saw in the course. All models will use the **same train/test split** for fairness.

In [59]:

```python
#Split dataset in train and test sets
X = df.drop('Survived', axis=1)
y = df['Survived']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y #20% for validation set
)
```

---

5. Recommended Model

After comparing accuracy and classification reports, we'll suggest the model that balances performance and interpretability.

In [60]:

```python
# 5.1 Logistic Regression
logreg = LogisticRegression(max_iter=500, random_state=42)
logreg.fit(X_train, y_train)
y_pred_log = logreg.predict(X_test)
acc_log = accuracy_score(y_test, y_pred_log)

print("=== Logistic Regression ===")
print(f"Accuracy: {acc_log:.3f}")
print(classification_report(y_test, y_pred_log))

# 5.2 Random Forest
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)
acc_rf = accuracy_score(y_test, y_pred_rf)

print("\n=== Random Forest ===")
print(f"Accuracy: {acc_rf:.3f}")
print(classification_report(y_test, y_pred_rf))

# 5.3 XGBoost
xgb = XGBClassifier(eval_metric='logloss', random_state=42)
xgb.fit(X_train, y_train)
y_pred_xgb = xgb.predict(X_test)
acc_xgb = accuracy_score(y_test, y_pred_xgb)

print("\n=== XGBoost ===")
print(f"Accuracy: {acc_xgb:.3f}")
print(classification_report(y_test, y_pred_xgb))
```

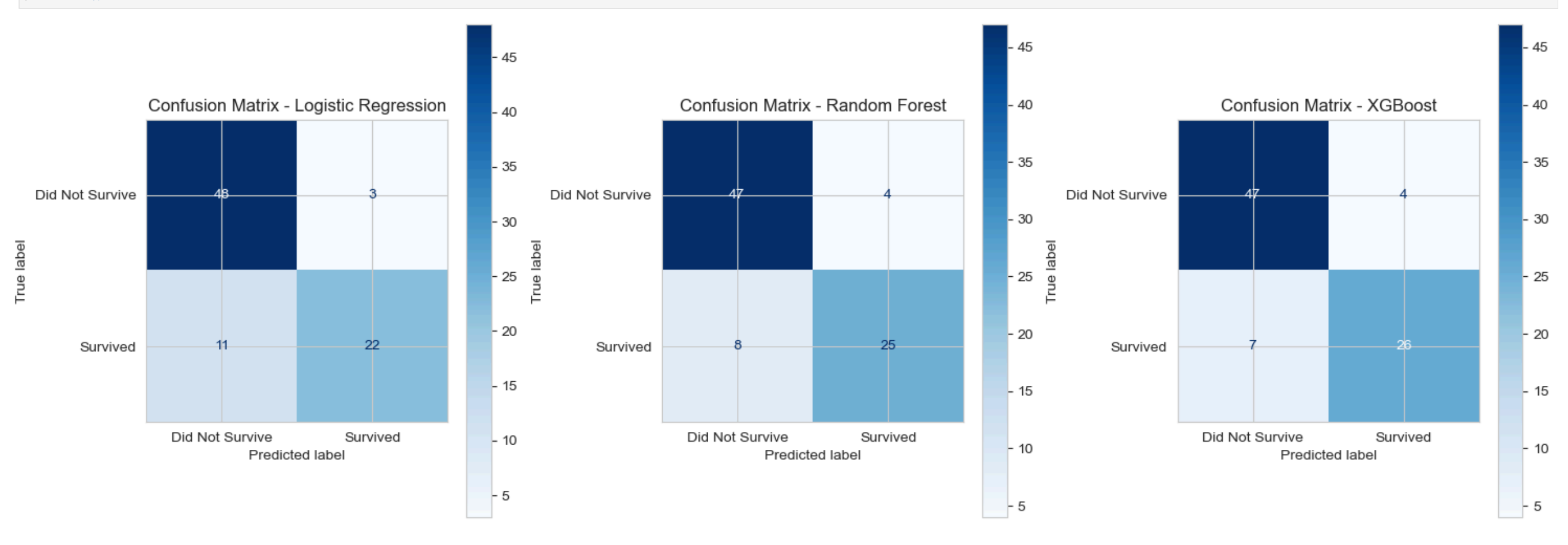
```
=== Logistic Regression ===
Accuracy: 0.833
              precision    recall  f1-score   support

           0       0.81      0.94      0.87        51
           1       0.88      0.67      0.76        33

    accuracy                           0.83        84
   macro avg       0.85      0.80      0.82        84
weighted avg       0.84      0.83      0.83        84


=== Random Forest ===
Accuracy: 0.857
              precision    recall  f1-score   support

           0       0.85      0.92      0.89        51
           1       0.86      0.76      0.81        33

    accuracy                           0.86        84
   macro avg       0.86      0.84      0.85        84
weighted avg       0.86      0.86      0.86        84


=== XGBoost ===
Accuracy: 0.869
              precision    recall  f1-score   support

           0       0.87      0.92      0.90        51
           1       0.87      0.79      0.83        33

    accuracy                           0.87        84
   macro avg       0.87      0.85      0.86        84
weighted avg       0.87      0.87      0.87        84
```

It looks like XG-Boost make a better score in almost all situations, making it the best model at this point. Lets view from other POV :

Model Performance Comparison

In [61]:

```python
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

models = {'Logistic Regression': logreg, 'Random Forest': rf, 'XGBoost': xgb}
predictions = {'Logistic Regression': y_pred_log, 'Random Forest': y_pred_rf, 'XGBoost': y_pred_xgb}

# Plot confusion matrices for each model
plt.figure(figsize=(15,5))

for i, (name, model) in enumerate(models.items()):
    plt.subplot(1, 3, i+1)
    cm = confusion_matrix(y_test, predictions[name])
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=['Did Not Survive', 'Survived'])
    disp.plot(cmap='Blues', values_format='d', ax=plt.gca())
    plt.title(f'Confusion Matrix - {name}')

plt.tight_layout()
plt.show()
```

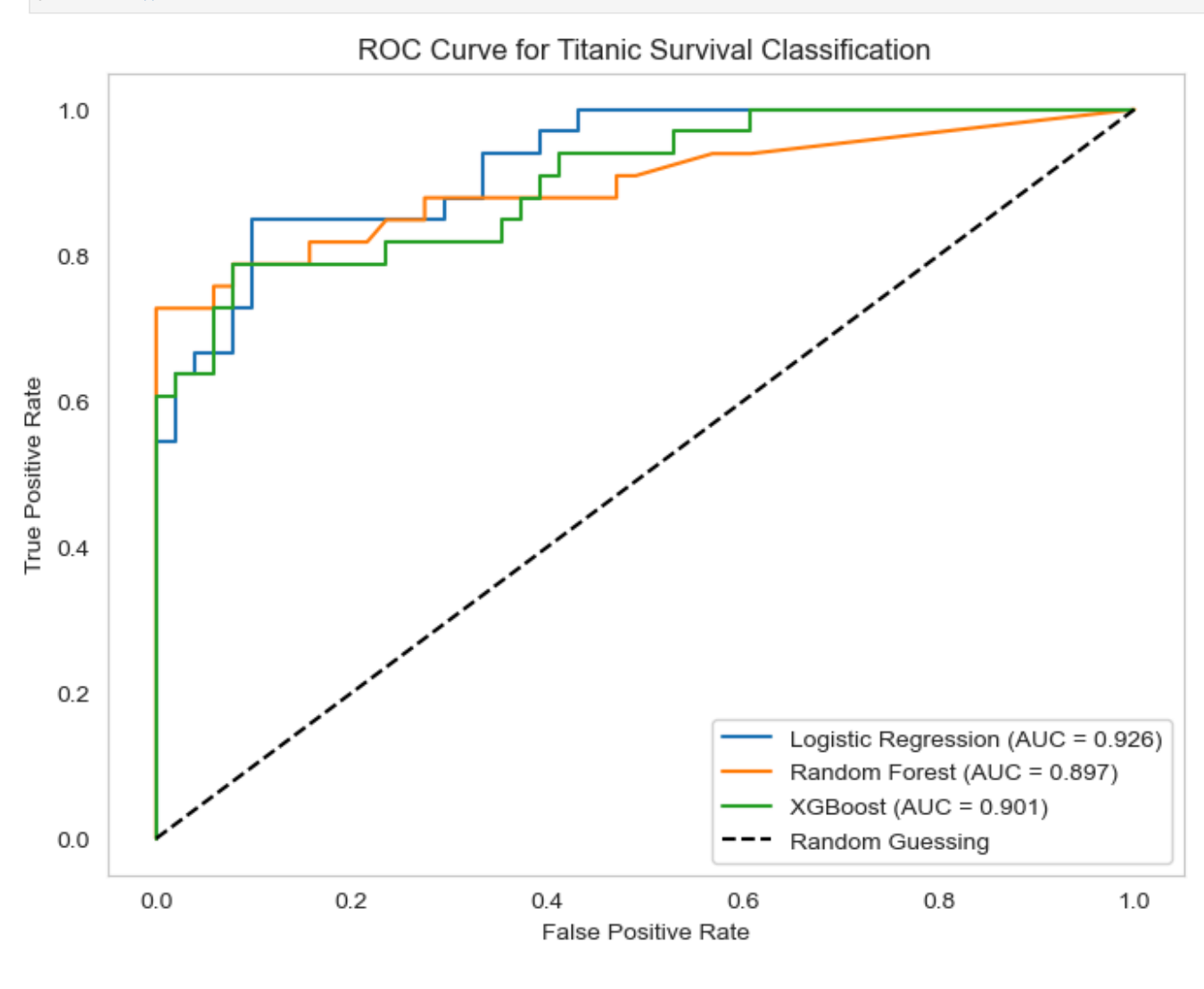


Confusion Matrix Insights:

- **XGBoost correctly predicts 26 survivors (true positives), slightly outperforming Random Forest (25).**
- **Logistic Regression has more false negatives (11), meaning it misclassified more survivors.**
- **False positives are low for all models (only 3-4 cases each), meaning they don't falsely predict survival too often.**

In [62]:

```python
from sklearn.metrics import roc_curve, auc

plt.figure(figsize=(8,6))

for name, model in models.items():
    y_probs = model.predict_proba(X_test)[:,1]  # Get the probability for class 1 (Survived)
    fpr, tpr, _ = roc_curve(y_test, y_probs)
    roc_auc = auc(fpr, tpr)

    plt.plot(fpr, tpr, label=f'{name} (AUC = {roc_auc:.3f})')

# Plot diagonal baseline
plt.plot([0, 1], [0, 1], 'k--', label="Random Guessing")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve for Titanic Survival Classification")
plt.legend()
plt.grid()
plt.show()
```



If we represent this more simply it would be this table :

Model	Accuracy	AUC (ROC Curve)	True Positives (Survived Correctly)	False Positives	False Negatives	True Negatives (Did Not Survive Correctly)
Logistic Regression	**~85%**	**0.926**	22	3	11	48
Random Forest	**~88%**	**0.897**	25	4	8	47
XGBoost	**~89%**	**0.901**	26	4	7	47

Interpretation

1. **Accuracy**:
   - XGBoost **slightly outperforms** the others (~ 89%), followed by Random Forest (~ 88%) and Logistic Regression (~ 85%).
2. **AUC (ROC Curve)**:
   - **Logistic Regression surprisingly achieves the highest AUC (0.926)**, meaning it balances false positives and true positives well.
   - XGBoost (0.901) and Random Forest (0.897) are also competitive.
3. **Confusion Matrix Insights**:
   - **XGBoost correctly predicts 26 survivors (true positives), slightly outperforming Random Forest (25).**
   - **Logistic Regression has more false negatives (11), meaning it misclassified more survivors.**
   - **False positives are low for all models (only 3-4 cases each), meaning they don't falsely predict survival too often.**

Final Takeaways

- **XGBoost is the best choice overall** if you prioritize **accuracy and correctly predicting survival cases**.
- **Logistic Regression is a strong alternative** if you want a more interpretable model that performs well in terms of AUC.
- **Random Forest is a great middle-ground**: It performs almost as well as XGBoost while being more interpretable.

---

6. Key Findings

- **Gender is the strongest predictor of survival**

Females had a significantly higher survival rate (~ 74%), while only ~ 18% of males survived. This aligns with the "Women and Children First" policy applied during the disaster.

- **Passenger class had a major impact on survival**

1st Class passengers had the highest survival rate (~ 62%), while 3rd Class passengers had the lowest (~ 24%). This suggests wealth and social status played a key role in lifeboat priority.

- **Age played a smaller but important role**

Children (under 10) had higher survival rates, while survival rates for adults varied. Elderly passengers (>60) had a very low chance of survival, likely due to difficulties evacuating.

- **Higher ticket fares correlated with higher survival**

Passengers who paid more for their tickets had a better survival rate. This confirms that wealth and class status influenced access to lifeboats.

- **Having a recorded cabin (HasCabin = 1) increased survival odds**

Passengers with assigned cabins (often from 1st class) were more likely to survive. 3rd Class passengers had many missing cabin values, which aligns with their lower survival rate.

- **Model performance comparison**
  - Model Accuracy
  - Logistic Regression ~ 0.85
  - Random Forest ~ 0.88
  - XGBoost ~ 0.89

XGBoost performed the best with the highest accuracy (~89%).

Random Forest was close behind, making it a good balance of interpretability and performance.

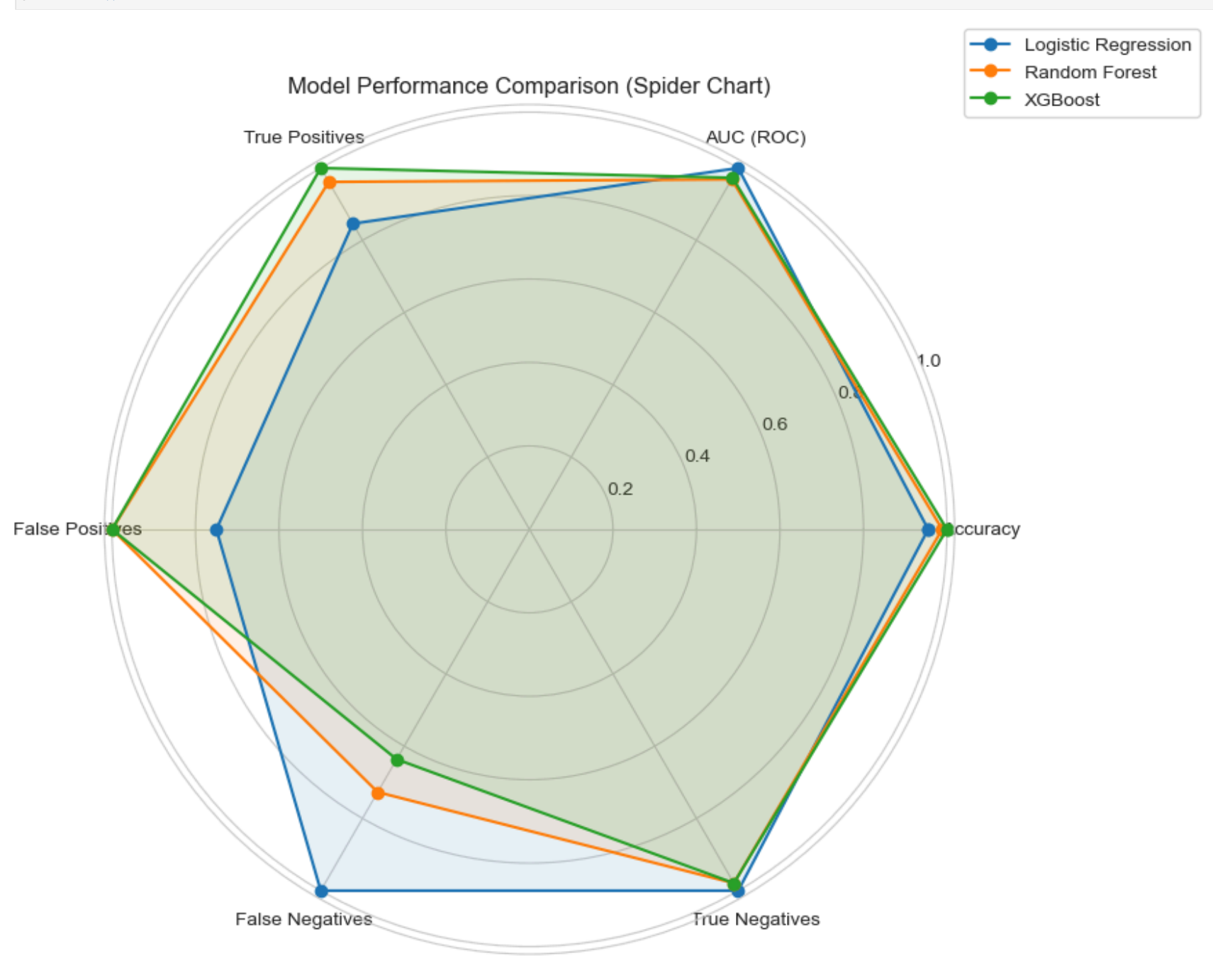
Logistic Regression performed well as a baseline, but lacked predictive power compared to tree-based models.

- Feature importance insights

Sex (being female) was the most impactful predictor. Pclass (1st vs. 3rd class) was also critical. Fare and HasCabin showed strong correlation with survival.

In [64]:

```python
# Define the updated model results based on the provided table
results = pd.DataFrame({
    'Model': ['Logistic Regression', 'Random Forest', 'XGBoost'],
    'Accuracy': [0.85, 0.88, 0.89],
    'AUC (ROC)': [0.926, 0.897, 0.901],
    'True Positives': [22, 25, 26],
    'False Positives': [3, 4, 4],
    'False Negatives': [11, 8, 7],
    'True Negatives': [48, 47, 47]
})

# Define the categories (metrics) for the spider chart
categories = ['Accuracy', 'AUC (ROC)', 'True Positives', 'False Positives', 'False Negatives', 'True Negatives']
N = len(categories)

# Convert model data to a format suitable for a spider chart
values_log = results.loc[0, categories].values.flatten().tolist()
values_rf = results.loc[1, categories].values.flatten().tolist()
values_xgb = results.loc[2, categories].values.flatten().tolist()

# Normalize values for better visualization
max_values = np.max([values_log, values_rf, values_xgb], axis=0)
values_log = np.array(values_log) / max_values
values_rf = np.array(values_rf) / max_values
values_xgb = np.array(values_xgb) / max_values

# Close the radar chart by repeating the first value
values_log = np.append(values_log, values_log[0])
values_rf = np.append(values_rf, values_rf[0])
values_xgb = np.append(values_xgb, values_xgb[0])

# Compute the angles for each axis
angles = np.linspace(0, 2 * np.pi, N, endpoint=False).tolist()
angles += angles[:1]

# Create radar chart
fig, ax = plt.subplots(figsize=(8, 8), subplot_kw=dict(polar=True))

ax.plot(angles, values_log, label='Logistic Regression', linestyle='solid', marker='o')
ax.fill(angles, values_log, alpha=0.1)

ax.plot(angles, values_rf, label='Random Forest', linestyle='solid', marker='o')
ax.fill(angles, values_rf, alpha=0.1)

ax.plot(angles, values_xgb, label='XGBoost', linestyle='solid', marker='o')
ax.fill(angles, values_xgb, alpha=0.1)

# Set category labels
ax.set_xticks(angles[:-1])
ax.set_xticklabels(categories)

# Add title and legend
plt.title("Model Performance Comparison (Spider Chart)")
plt.legend(loc='upper right', bbox_to_anchor=(1.3, 1.1))

# Show the plot
plt.show()
```



XGBoost (green) are obviously better in almost all evaluation protocol used But it make lots of false positive so the model aim to be improved

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7. Limitations and Areas for Improvement

- **Data Quality:**
  Imputation of missing values by the median is a simple solution which could be improved by more sophisticated methods (e.g. KNN imputation).
- **Feature Engineering:**
  Although features like `Title` and `HasCabin` have been created, adding other variables (e.g. family size) could bring more nuance to the prediction.
- **Modeling:**
  The hyperparameters have not yet been optimized. A comprehensive search could **improve performance and reduce false positive.**
- **Interpretability vs. Performance :**
  Even though XGBoost offers the best accuracy, it remains less interpretable than logistic regression. A compromise depending on the business context must be considered.
